

Supplementary Material

A Framework for Efficient Search of High-Distance Quantum Stabilizer Codes

1 Search-Space Retention Factors for Structural Constraints and Other Calculations

This section provides the calculations of the search-space retention factor for the structural constraints considered in the main text.

1.1 Universal Stabilizer Constraints

Let $r = n - k_n$. The unrestricted search space of $r \times 2n$ matrices over $GF(q)$ is

$$S_O = q^{2nr}.$$

Before imposing commutation, the number of full-row-rank matrices is

$$S_{FR} = \prod_{i=1}^r (q^{2n} - q^{i-1}) = q^{2nr} \prod_{i=1}^r (1 - q^{-2n+i-1}).$$

For $r = \Theta(n)$, the product contributes only $O(1)$ to $\log_q S_{FR}$, and therefore does not affect the leading $O(nr)$ exponent.

Under the symplectic commutation constraint

$$H_X H_Z^T = H_Z H_X^T,$$

the rows of H form an ordered basis of a totally isotropic subspace of $GF(q)^{2n}$. Sequentially, after $i - 1$ linearly independent commuting rows have been chosen, the i -th row can be selected from $W_{i-1}^\perp \setminus W_{i-1}$, giving

$$q^{2n-i+1} - q^{i-1}$$

choices. Hence,

$$S_R = \prod_{i=1}^r (q^{2n-i+1} - q^{i-1}) = q^{\binom{r}{2}} \prod_{j=n-r+1}^n (q^{2j} - 1).$$

Factoring q^{2j} from each term gives

$$S_R = q^{2nr - \binom{r}{2}} \prod_{j=n-r+1}^n (1 - q^{-2j}).$$

Thus,

$$\log_{q^2} S_R = nr - \frac{r^2}{4} + \frac{r}{4} + O(1).$$

For a code family satisfying

$$r = n - k_n, \quad \lim_{n \rightarrow \infty} \frac{k_n}{n} = R,$$

we have $r/n \rightarrow 1 - R$. Therefore, the search-space retention factor is

$$\mathcal{R}_{[[n, k_n, d]]_q} = \lim_{n \rightarrow \infty} \frac{\log_{q^2} S_R}{\log_{q^2} S_O} = 1 - \frac{1 - R}{4} = \frac{3 + R}{4}.$$

Note: R denotes the code rate here; in the main paper, the code rate is denoted by r .

1.2 Weight Spectrum

The number of possible configurations of a row of weight w_i is

$$\binom{n}{w_i} (q^2 - 1)^{w_i}.$$

Allowing the row weight to take any value in $\mathcal{W} = \{w_1, \dots, w_m\}$, the number of configurations is

$$\sum_{i=1}^m \binom{n}{w_i} (q^2 - 1)^{w_i}.$$

Since the $n - k_n$ rows are independent, the reduced search space is

$$S_R = \left(\sum_{i=1}^m \binom{n}{w_i} (q^2 - 1)^{w_i} \right)^{n - k_n},$$

while, by default,

$$S_O = q^{2n(n - k_n)}.$$

Hence,

$$\frac{\log_{q^2} S_R}{\log_{q^2} S_O} = \frac{1}{n} \log_{q^2} \left(\sum_{i=1}^m \binom{n}{w_i} (q^2 - 1)^{w_i} \right).$$

Let

$$\delta = \lim_{n \rightarrow \infty} \frac{w_{\max}}{n}, \quad w_{\max} = \max(\mathcal{W}).$$

Since the sum contains finitely many positive terms, its exponential growth rate is determined by its largest term. Therefore,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log_{q^2} \left(\sum_{i=1}^m \binom{n}{w_i} (q^2 - 1)^{w_i} \right) = \lim_{n \rightarrow \infty} \frac{1}{n} \log_{q^2} \left[\binom{n}{\delta n} (q^2 - 1)^{\delta n} \right].$$

Using Stirling's approximation,

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log_{q^2} \binom{n}{\delta n} = -\delta \log_{q^2} \delta - (1 - \delta) \log_{q^2} (1 - \delta),$$

and

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log_{q^2} (q^2 - 1)^{\delta n} = \delta \log_{q^2} (q^2 - 1).$$

Thus,

$$\lim_{n \rightarrow \infty} \frac{\log_{q^2} S_R}{\log_{q^2} S_O} = \delta \log_{q^2} (q^2 - 1) - \delta \log_{q^2} \delta - (1 - \delta) \log_{q^2} (1 - \delta),$$

which is precisely

$$\lim_{n \rightarrow \infty} \frac{\log_{q^2} S_R}{\log_{q^2} S_O} = H_{q^2}(\delta).$$

1.3 Qudit Participation

Regular Qudit Participation

The search-space calculation for regular qudit participation is equivalent to the weight-spectrum calculation under the substitutions

$$n \rightarrow n - k_n, \quad n - k_n \rightarrow n.$$

Thus, requiring every physical qudit to participate in exactly r stabilizer generators gives

$$S_R = \left[\binom{n - k_n}{r} (q^2 - 1)^r \right]^n,$$

and hence

$$\mathcal{R}_{[[n, k_n, d]]_q} = H_{q^2}(\delta'), \quad \delta' = \lim_{n \rightarrow \infty} \frac{r}{n - k_n}.$$

Restricted Qudit Participation

Similarly, the search-space calculation for restricted qudit participation follows from the weight-spectrum calculation under

$$n \rightarrow n - k_n, \quad n - k_n \rightarrow n.$$

For the restriction

$$Q(G|a)) \leq r_{\max}, \quad \forall a \in \{0, 1, \dots, n - 1\},$$

the reduced search space is

$$S_R = \left(\sum_{i=0}^{r_{\max}} \binom{n - k_n}{i} (q^2 - 1)^i \right)^n,$$

with

$$\mathcal{R}_{[[n, k_n, d]]_q} = H_{q^2}(\delta'), \quad \delta' = \lim_{n \rightarrow \infty} \frac{r_{\max}}{n - k_n}.$$

1.4 Overlap Constraints

p -wise Overlap Constraints

Let M be an $r \times n$ matrix over $GF(q^2)$, where $r = n - k_n$, and let $S \in \{0, 1\}^{r \times n}$ denote its binary support matrix, with $S_{ij} = 1$ if $M_{ij} \neq 0$. For a support matrix S with total weight $W(S)$, the number of corresponding $GF(q^2)$ assignments is

$$(q^2 - 1)^{W(S)}.$$

The p -wise overlap of rows $\{i_1, \dots, i_p\}$ is

$$I_p(i_1, \dots, i_p) = \sum_{j=1}^n S_{i_1 j} S_{i_2 j} \cdots S_{i_p j}.$$

We assume

$$\delta_p = \lim_{n \rightarrow \infty} \frac{\lambda_p}{n}, \quad \delta_\mu = \lim_{n \rightarrow \infty} \frac{\mu_p}{n}, \quad r = \Theta(n).$$

Case I: Symmetric p -wise Overlap

For the constraint

$$I_p = \lambda_p \quad \text{for all distinct } p\text{-tuples of rows,}$$

the columns of S form the incidence structure of a p -design. For $p \geq 4$, the generalized Fisher inequality implies that a non-trivial design requires

$$n \geq \binom{r}{\lfloor p/2 \rfloor},$$

which is asymptotically incompatible with $r = \Theta(n)$. Hence, for $p \geq 4$, the constraint set is empty and $S_R = 0$.

For $p \in \{2, 3\}$, valid designs may exist, but their number N_{design} has negligible entropy relative to the q^{2rn} search space:

$$\log N_{\text{design}} = o(rn).$$

The required column density is

$$\rho = \delta_p^{1/p},$$

giving total support weight

$$W = rn\delta_p^{1/p}.$$

Thus,

$$S_R \sim N_{\text{design}}(q^2 - 1)^{rn\delta_p^{1/p}},$$

and therefore

$$\mathcal{R}_{[[n, k_n, d]]_q} = \delta_p^{1/p} \log_{q^2}(q^2 - 1).$$

Case II: Quasi-symmetric p -wise Overlap

For the constraint

$$I_p \in \{\lambda_p, \mu_p\},$$

define

$$\delta_{\max} = \max(\delta_p, \delta_\mu).$$

The corresponding support structures form a two-intersection p -design. As in the symmetric case, their number $N_{\text{q-design}}$ has negligible asymptotic entropy:

$$\log N_{\text{q-design}} = o(rn).$$

Maximizing the number of Pauli assignments therefore gives the effective support density

$$\rho = \delta_{\max}^{1/p},$$

and hence

$$S_R \sim N_{\text{q-design}}(q^2 - 1)^{rn\delta_{\max}^{1/p}}.$$

Consequently,

$$\mathcal{R}_{[[n, k_n, d]]_q} = \delta_{\max}^{1/p} \log_{q^2}(q^2 - 1).$$

The symmetric and quasi-symmetric cases therefore have the same asymptotic form when evaluated at the corresponding maximum overlap, while their applicability remains subject to the existence of the required combinatorial designs.

Case III: Bounded p -wise Overlap

Consider the bounded p -wise overlap constraint

$$I_p(i_1, \dots, i_p) \leq \lambda_p \quad \text{for all distinct } p\text{-tuples of rows.}$$

Let $M \in GF(q^2)^{r \times n}$, where $r = n - k_n$, and let $S \in \{0, 1\}^{r \times n}$ denote its support matrix. For the j -th column, let w_j denote its Hamming weight. A support matrix with column weights $\{w_j\}_{j=1}^n$ corresponds to

$$\prod_{j=1}^n \binom{r}{w_j} (q^2 - 1)^{w_j}$$

distinct $GF(q^2)$ matrices.

Summing the p -wise overlaps over all p -subsets of rows gives the exact identity

$$\sum_{|A|=p} I_p(A) = \sum_{j=1}^n \binom{w_j}{p}.$$

Since $I_p(A) \leq \lambda_p$ for every A ,

$$\sum_{j=1}^n \binom{w_j}{p} \leq \binom{r}{p} \lambda_p.$$

Define

$$\delta_p = \lim_{n \rightarrow \infty} \frac{\lambda_p}{n}, \quad \rho_j = \frac{w_j}{r}.$$

Since $r = \Theta(n)$, the preceding constraint becomes asymptotically

$$\frac{1}{n} \sum_{j=1}^n \rho_j^p \leq \delta_p.$$

For a fixed column density ρ_j , the logarithmic number of $GF(q^2)$ assignments is asymptotically

$$r H_{q^2}(\rho_j),$$

where

$$H_{q^2}(\rho) = -\rho \log_{q^2} \rho - (1 - \rho) \log_{q^2} (1 - \rho) + \rho \log_{q^2} (q^2 - 1).$$

For

$$\rho \leq 1 - \frac{1}{q^2},$$

$H_{q^2}(\rho)$ is increasing. Setting $z_j = \rho_j^p$, the function $H_{q^2}(z_j^{1/p})$ is concave on this range. Hence, by Jensen's inequality, the asymptotic entropy is maximized by a uniform column density,

$$\rho_j = \rho = \delta_p^{1/p}.$$

If this exceeds the unconstrained density $1 - q^{-2}$, the entropy is instead maximized at that natural density. Thus,

$$\rho^* = \min \left(\delta_p^{1/p}, 1 - \frac{1}{q^2} \right).$$

The corresponding asymptotic search-space volume is

$$S_R = (q^2)^{rnH_{q^2}(\rho^*) + o(rn)}.$$

Since

$$S_O = (q^2)^{rn},$$

the search-space retention factor is

$$\mathcal{R}_{[[n, k_n, d]]_q} = H_{q^2}(\rho^*).$$

The upper bound is asymptotically achievable by choosing the column density slightly below ρ^* . Specifically, taking $\rho_n = \rho^* - o(1)$ makes the expected p -wise overlap strictly smaller than λ_p , while concentration inequalities and a union bound over the $\binom{r}{p} = O(n^p)$ row tuples ensure that all overlap constraints are satisfied with probability tending to one. Since $\rho_n \rightarrow \rho^*$, this vanishing back-off does not alter the limiting retention factor.

Therefore,

$$\mathcal{R}_{[[n, k_n, d]]_q} = H_{q^2} \left(\min \left\{ \delta_p^{1/p}, 1 - \frac{1}{q^2} \right\} \right).$$

For the non-vacuous regime $\delta_p^{1/p} \leq 1 - q^{-2}$, this is equivalently

$$\mathcal{R}_{[[n, k_n, d]]_q} = \frac{H_q(\delta_p^{1/p}) + \delta_p^{1/p} \log_q(q+1)}{2}.$$

In the complementary regime, $\delta_p^{1/p} \geq 1 - q^{-2}$, the overlap constraint is asymptotically vacuous, since the unconstrained q^2 -ary entropy is maximized at $\rho = 1 - q^{-2}$. Consequently,

$$\mathcal{R}_{[[n, k_n, d]]_q} = 1.$$

For $p = 2$, the bounded p -wise overlap constraint reduces to the pairwise overlap constraint represented by the Gram matrix. Thus, the corresponding retention factor reduces to the Gram-matrix case as a special case of the general p -wise overlap formulation.³

1.5 Support-Containment Constraint

Let $r = n - k_n$, and consider the constraint

$$\text{supp}(g_i) \not\subseteq \text{supp}(g_j), \quad \forall i \neq j.$$

The unrestricted search space is

$$S_O = q^{2nr}.$$

For a pair of rows (i, j) , let E_{ij} denote the set of configurations satisfying $\text{supp}(g_i) \subseteq \text{supp}(g_j)$. For a single column, the allowed support patterns are $(0, 0)$, $(0, 1)$, $(1, 1)$, with respective numbers of Pauli assignments 1, $q^2 - 1$, and $(q^2 - 1)^2$. Hence, the number of configurations for the pair is

$$[1 + (q^2 - 1) + (q^2 - 1)^2]^n = (q^4 - q^2 + 1)^n.$$

The remaining $r - 2$ rows are unrestricted, giving

$$|E_{ij}| = (q^2)^{n(r-2)}(q^4 - q^2 + 1)^n.$$

There are $r(r-1)$ ordered pairs, so the union bound gives

$$S_R \geq S_O - r(r-1)(q^2)^{n(r-2)}(q^4 - q^2 + 1)^n.$$

Equivalently,

$$S_R \geq S_O \left[1 - r(r-1) \left(1 - \frac{q^2 - 1}{q^4} \right)^n \right].$$

Since $r \leq n$ and

$$0 < 1 - \frac{q^2 - 1}{q^4} < 1,$$

the correction term vanishes exponentially with n . Thus,

$$S_R = S_O \left[1 - O \left(n^2 \left(1 - \frac{q^2 - 1}{q^4} \right)^n \right) \right].$$

Consequently,

$$\log_{q^2} S_R = n(n - k_n) - o(n(n - k_n)),$$

and therefore

$$\mathcal{R}_{[[n, k_n, d]]_q} = \lim_{n \rightarrow \infty} \frac{\log_{q^2} S_R}{\log_{q^2} S_O} = 1.$$

Hence, the support-containment constraint does not reduce the search space at the leading exponential order. The result holds for $r = \Theta(n)$, and more generally whenever r grows at most polynomially with n .

1.6 Pauli-Distribution Constraints

Generator-Wise Equal Distribution

For the generalized Pauli alphabet over $GF(q^2)$, partition the $q^2 - 1$ non-identity elements into $q^2 - 1$ equally sized classes and impose equal distribution among these classes. Let t_i denote the number of occurrences of each non-identity class in generator g_i . Thus, for each i ,

$$N_1(g_i) = N_2(g_i) = \dots = N_{q^2-1}(g_i) = t_i.$$

For a fixed value $t_i = t$, the number of valid length- n strings is

$$\binom{n}{\underbrace{t, t, \dots, t}_{q^2-1 \text{ times}}, n - (q^2 - 1)t} = \frac{n!}{(t!)^{q^2-1} (n - (q^2 - 1)t)!}.$$

For $q = 2$, there are $q^2 - 1 = 3$ non-identity Pauli operators, corresponding to X, Y, Z , and the constraint reduces to

$$N_X(g_i) = N_Y(g_i) = N_Z(g_i) = t_i.$$

34 The exact single-generator count over $GF(q^2)$ is given by

$$A_n(q) = \sum_{t=0}^{\lfloor n/(q^2-1) \rfloor} \frac{n!}{(t!)^{q^2-1} (n - (q^2 - 1)t)!}.$$

Since the constraint is imposed independently on each of the $r = n - k_n$ generators,

$$S_R = (A_n(q))^{n-k_n}.$$

The dominant term in the sum occurs at $t/n \rightarrow 1/q^2$. Applying Stirling's approximation around this peak yields the asymptotic growth

$$A_n(q) \sim \frac{q^{2n}}{\sqrt{q^2 - 1}} \left(\frac{q^2}{2\pi n} \right)^{\frac{q^2 - 2}{2}}.$$

which for the qubit case ($q = 2$) simplifies to $A_n(2) \sim \frac{4^{n \cdot 2}}{\pi \sqrt{3} n} .5453$

Taking the logarithmic rate gives

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log_{q^2} A_n(q) = 1.$$

Since $S_O = q^{2n(n-k_n)}$, the overall search-space retention factor is

$$\mathcal{R}_{[[n, k_n, d]]_q} = \lim_{n \rightarrow \infty} \frac{\log_{q^2} S_R}{\log_{q^2} S_O} = \lim_{n \rightarrow \infty} \frac{(n - k_n) \log_{q^2} A_n(q)}{n(n - k_n)} = 1.$$

Weaker Pauli Distribution Constraints

The constraints of globally equal distribution, quasi-equal distribution, and restriction on variance are weaker than the generator-wise equal distribution constraint. Since the generator-wise constraint already has asymptotic retention factor 1, these weaker constraints necessarily satisfy

$$\mathcal{R}_{[[n, k_n, d]]_q} = 1$$

as well.

1.7 Mixed Constraints

Let S_O denote the original search space and let $S_{R_i} \subseteq S_O$ be the subset satisfying constraint i . The space satisfying all m constraints simultaneously is

$$S_{R_{\text{eff}}} = \bigcap_{i=1}^m S_{R_i}.$$

For each constraint, define

$$\mathcal{R}_i = \frac{\log_{q^2} S_{R_i}}{\log_{q^2} S_O} = \log_{S_O} S_{R_i},$$

so that

$$S_{R_i} = S_O^{\mathcal{R}_i}.$$

Since $S_{R_{\text{eff}}} \subseteq S_{R_i}$ for every i ,

$$\mathcal{R}_{\text{eff}} \leq \min_{1 \leq i \leq m} \mathcal{R}_i.$$

For the lower bound, let

$$A_i = S_O \setminus S_{R_i}$$

denote the configurations excluded by constraint i . By the union bound,

$$\left| \bigcup_{i=1}^m A_i \right| \leq \sum_{i=1}^m |A_i|,$$

and hence

$$S_{R_{\text{eff}}} \geq \sum_{i=1}^m S_{R_i} - (m-1)S_O.$$

Substituting $S_{R_i} = S_O^{\mathcal{R}_i}$ gives

$$S_{R_{\text{eff}}} \geq \sum_{i=1}^m S_O^{\mathcal{R}_i} - (m-1)S_O.$$

Writing the resulting lower bound in the form

$$\mathcal{R}_{\text{eff}} \geq \prod_{i=1}^m \mathcal{R}_i - d,$$

the corresponding deviation is

$$d = \log_{S_O} S,$$

where

$$S = \frac{S_O^{\prod_{i=1}^m \mathcal{R}_i}}{\sum_{i=1}^m S_O^{\mathcal{R}_i} - (m-1)S_O}.$$

To verify that $S \geq 1$, let

$$B = S_O > 1, \quad x_i = \mathcal{R}_i \in [0, 1].$$

It suffices to show

$$B^{\prod_{i=1}^m x_i} + (m-1)B \geq \sum_{i=1}^m B^{x_i}.$$

For $m = 2$, define

$$f(x_1) = B^{x_1 x_2} - B^{x_1}.$$

Since $x_2 \leq 1$ and $x_1 x_2 \leq x_1$,

$$f'(x_1) = \ln(B) (x_2 B^{x_1 x_2} - B^{x_1}) \leq 0.$$

Thus f is decreasing, and

$$f(x_1) \geq f(1) = B^{x_2} - B,$$

which gives

$$B^{x_1 x_2} + B \geq B^{x_1} + B^{x_2}.$$

The result extends inductively to arbitrary m . If it holds for $m = k-1$, set

$$Y = \prod_{i=1}^{k-1} x_i \in [0, 1].$$

Applying the $m = 2$ result to Y and x_k gives

$$B^{Y x_k} + B \geq B^Y + B^{x_k}.$$

Adding $(k-2)B$ and applying the induction hypothesis yields

$$B \prod_{i=1}^k x_i + (k-1)B \geq \sum_{i=1}^k B^{x_i}.$$

Hence, for all $m \geq 2$,

$$S_O^{\prod_{i=1}^m \mathcal{R}_i} \geq \sum_{i=1}^m S_O^{\mathcal{R}_i} - (m-1)S_O,$$

and therefore

$$S \geq 1.$$

To ensure a non-negative effective retention factor, d is capped by the product of the individual retention factors:

$$d = \min \left\{ \prod_{i=1}^m \mathcal{R}_i, \log_{S_O} S \right\}.$$

Thus,

$$\prod_{i=1}^m \mathcal{R}_i - d \leq \mathcal{R}_{\text{eff}} \leq \min_{1 \leq i \leq m} \mathcal{R}_i.$$

Mutually Independent Constraints

For two constraints, logarithmic independence requires that applying the second constraint to the space already reduced by the first retains the same logarithmic fraction as it does on the original space:

$$\frac{\log S_{\text{R}_{\text{eff}}}}{\log S_{R_1}} = \frac{\log S_{R_2}}{\log S_O}.$$

Consequently,

$$\mathcal{R}_{\text{eff}} = \mathcal{R}_1 \mathcal{R}_2.$$

More generally, let

$$S^{(k)} = \bigcap_{i=1}^k S_{R_i}.$$

If the constraints are mutually logarithmically independent, then

$$\frac{\log S^{(k)}}{\log S^{(k-1)}} = \mathcal{R}_k, \quad k = 1, \dots, m,$$

with $S^{(0)} = S_O$. Hence,

$$\log S_{\text{R}_{\text{eff}}} = \left(\prod_{i=1}^m \mathcal{R}_i \right) \log S_O,$$

and therefore

$$\mathcal{R}_{\text{eff}} = \prod_{i=1}^m \mathcal{R}_i.$$

Pairwise independence alone is insufficient for the general m -constraint product; mutual sequential logarithmic independence is required.

1.8 Other Calculations

1.8.1 Local Non-Commutation Count

An element of a Galois string over $GF(q^2)$ represents a Pauli operator acting on a single qudit. Two single-qudit Pauli operators commute if and only if they are equal or at least one of them is the identity. In the $GF(q^2)$ representation, this condition is equivalent to

$$ab(a + b) = 0.$$

Thus, for two Pauli operators P_1 and P_2 of length n , the local non-commutation count is defined as

$$C(P_1, P_2) = \sum_{i=1}^n \chi((\alpha_1)_i, (\alpha_2)_i),$$

where

$$\chi(a, b) = \begin{cases} 0, & (a + b)ab = 0, \\ 1, & (a + b)ab \neq 0. \end{cases}$$

The quantity $C(P_1, P_2)$ therefore counts the number of qudits on which the corresponding single-qudit Pauli operators do not commute.

1.8.2 Finite-Size Correction for Weight-Spectrum Constraints

Let $r = n - k_n$. For a prescribed set of generator weights $\mathcal{W} = \{w_1, \dots, w_m\}$, the restricted search space is

$$S_R = X^r, \quad X = \sum_{i=1}^m \binom{n}{w_i} (q^2 - 1)^{w_i},$$

while the unrestricted search space is

$$S_O = (q^2)^{nr}.$$

Hence, the finite- n retention factor is

$$R_n = \frac{\log_{q^2} S_R}{\log_{q^2} S_O} = \frac{1}{n} \log_{q^2} X.$$

Let

$$w_{\max} = \max(\mathcal{W}), \quad \delta_{\max} = \frac{w_{\max}}{n},$$

and define the dominant contribution

$$X_{\text{dom}} = \binom{n}{w_{\max}} (q^2 - 1)^{w_{\max}}.$$

Then

$$\log_{q^2} X = \log_{q^2} X_{\text{dom}} + \log_{q^2} \left(\frac{X}{X_{\text{dom}}} \right).$$

Using Stirling's approximation,

$$\log_{q^2} X_{\text{dom}} = n H_{q^2}(\delta_{\max}) - \frac{1}{2} \log_{q^2} [2\pi n \delta_{\max} (1 - \delta_{\max})] + o(1).$$

Therefore,

$$R_n = H_{q^2}(\delta_{\max}) - \frac{\log_{q^2} [2\pi n \delta_{\max}(1 - \delta_{\max})]}{2n} + \frac{1}{n} \log_{q^2} \left(\frac{X}{X_{\text{dom}}} \right) + o\left(\frac{1}{n}\right).$$

Since the asymptotic retention factor is

$$\mathcal{R} = H_{q^2}(\delta_{\max}),$$

the finite-size correction

$$C_n = \mathcal{R} - R_n$$

is

$$C_n = \frac{\log_{q^2} [2\pi n \delta_{\max}(1 - \delta_{\max})]}{2n} - \frac{1}{n} \log_{q^2} \left(\frac{X}{X_{\text{dom}}} \right) + o\left(\frac{1}{n}\right).$$

Thus, $C_n = O(\log n/n)$, and consequently

$$\lim_{n \rightarrow \infty} C_n = 0.$$

1.8.3 Implications for Unresolved Semi-Perfect Quantum Codes

The estimation of C' is carried out empirically, for which the corresponding code has been provided. The weight spectrum, without frequencies, is estimated empirically and manually by analyzing the neighboring codes. Using the estimated weight spectrum, S_R is calculated as

$$S_R = \left(\sum_{i=1}^m \binom{n}{w_i} (q^2 - 1)^{w_i} \right)^{n-k_n}.$$

We then calculate $\mathcal{R}$ using S_R . Using the empirically estimated finite-size correction factor C' , we obtain

$$H_{q^2}(\delta_{\text{avg}}) = \mathcal{R} + C', \quad \delta_{\text{avg}} = \frac{w_{\text{avg}}}{n},$$

from which w_{avg} is determined.

Using the estimated w_{avg} , the weight spectrum without frequencies, and $n - k_n$, we determine the possible values of the frequencies f_i , from which the most plausible frequency distribution is selected manually.

With the frequencies f_i now estimated, the theoretical search-space size is calculated as

$$S_R = \prod_i \left[\binom{n}{w_i} (q^2 - 1)^{w_i} \right]^{f_i},$$

and the corresponding theoretical runtime speedup is

$$\text{Theoretical Runtime Speedup} = \frac{S_O}{S_R}.$$

Note: Using the method described in Section 5.1 of the main paper, the reduced search space is given by

$$S_R = \sum_i \left[\binom{n}{w_i} (q^2 - 1)^{w_i} \right]^{f_i},$$

resulting in a substantial reduction in the search space and a corresponding reduction in computational cost.

1.8.4 Performance Under Stochastic Noise

The probability of an error of weight w occurring in a codeword of length n under an independent stochastic noise model is given by

$$P(w) = \binom{n}{w} p^w (1-p)^{n-w},$$

where p is the probability of an error occurring on a single qudit. If only a of the $\binom{n}{w}(q^2-1)^w$ possible errors of weight w are correctable or detectable, the probability of a correctable or detectable error of weight w is

$$P_{\text{correctable/detectable}}(w) = \frac{a}{\binom{n}{w}(q^2-1)^w} P(w).$$

Thus, the cumulative probability of a correctable or detectable error of weight less than or equal to w occurring is

$$\sum_{i=1}^w P_{\text{correctable/detectable}}(i)$$

The probability that an error of weight less than or equal to w is correctable or detectable is given by the probability of a correctable or detectable error of weight less than or equal to w occurring, divided by the probability of an error of weight less than or equal to w occurring. Thus,

$$P_{\text{correct/detect}}(p) = \frac{\sum_{i=1}^w P_{\text{correctable/detectable}}(i)}{\sum_{i=1}^w P(i)}$$

This demonstrates how closely the presented code is related to the theoretically optimal $[[14, 3, 5]]_2$ code.

2 Additional Plots

The following figures provide additional analysis supporting the correction factors considered in the final analysis. During preliminary exploration, three candidate correction factors were investigated. The first and third were subsequently retained as C' and C , respectively, while the second exhibited substantially weaker empirical correlation and was excluded from the final analysis. The figures below therefore concern only the retained correction factors C and C' .

2.1 C versus k/n for all codes

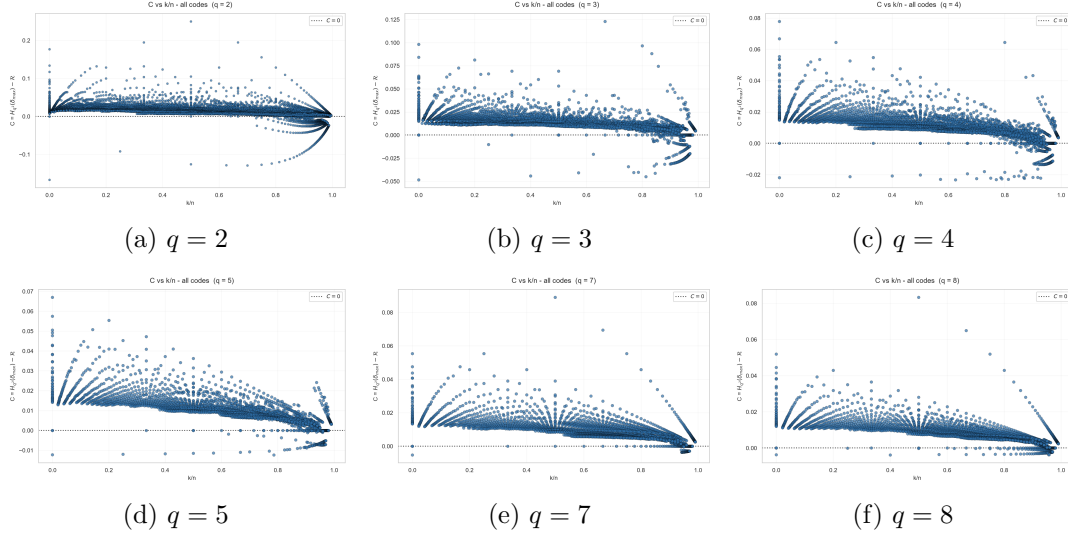


Figure 2.1: Finite-size correction C as a function of the code rate k/n for quantum codes across field sizes $q = 2, 3, 4, 5, 7, 8$.

2.2 C' versus k/n for semi-perfect quantum codes

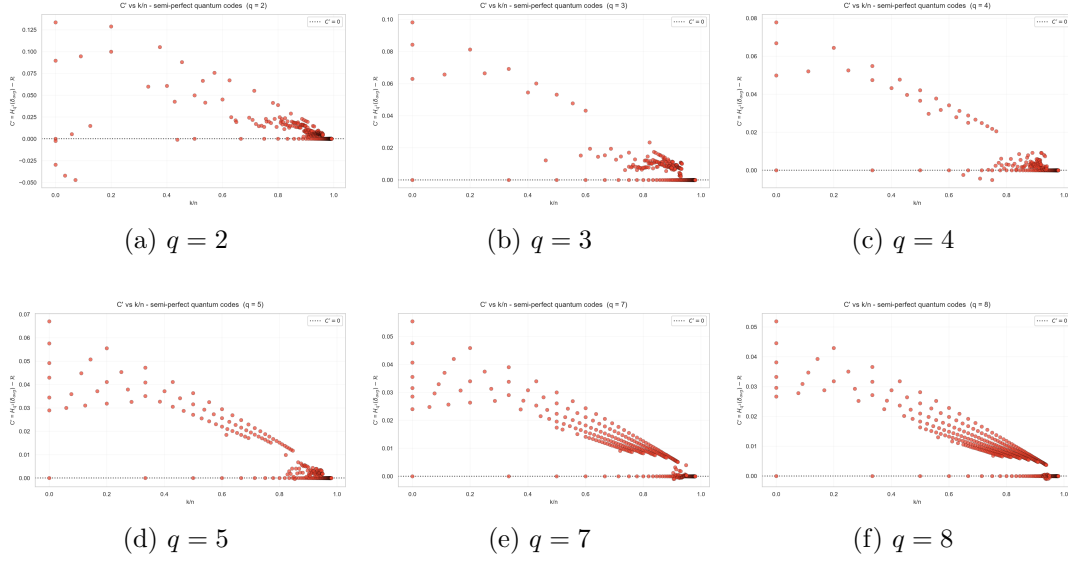


Figure 2.2: Modified correction factor C' versus code rate k/n for semi-perfect quantum codes across field sizes $q = 2, 3, 4, 5, 7, 8$.

2.3 Finite-Size Scaling of the Correction Factor

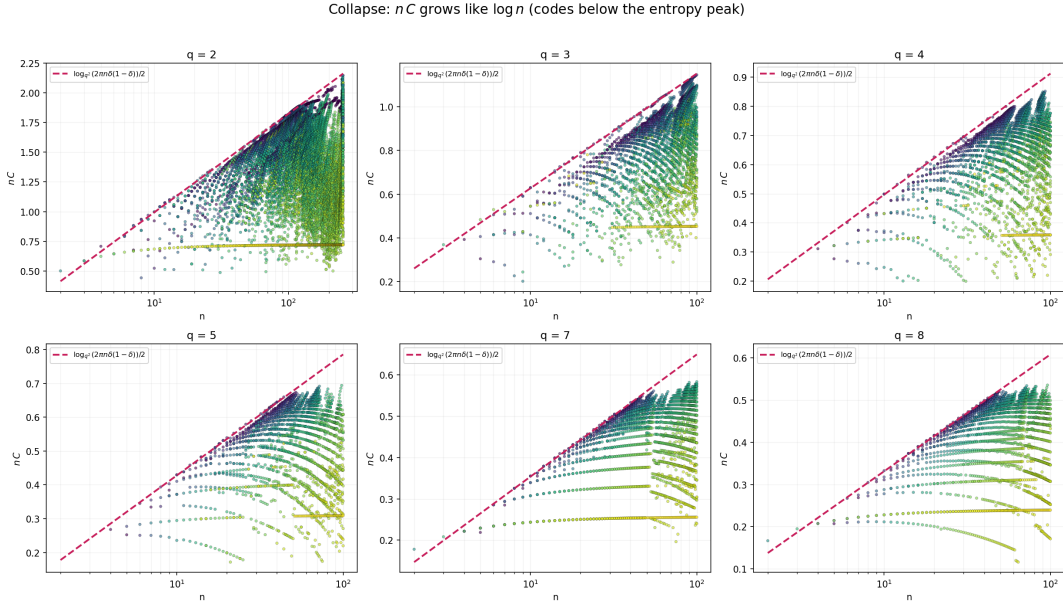


Figure 2.3: Scaling of nC with $\log n$ for codes below the entropy peak across field sizes $q = 2, 3, 4, 5, 7, 8$.

2.4 Empirical Estimation of the C' Correction Factor

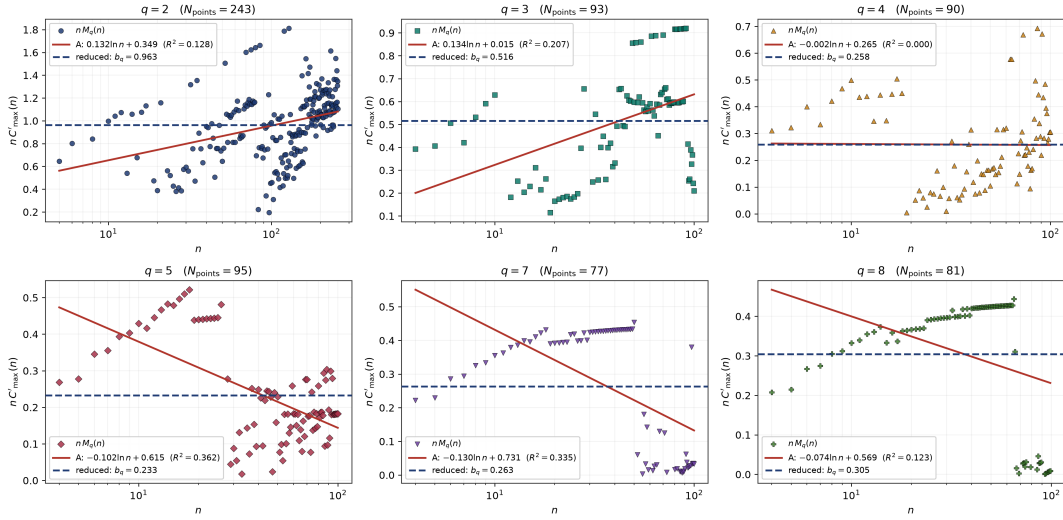


Figure 2.4: Upper envelope of the modified correction factor C' in collapse coordinates for semi-perfect quantum codes across field sizes $q = 2, 3, 4, 5, 7, 8$.

2.5 Comparison of the C' and C Correction Factors

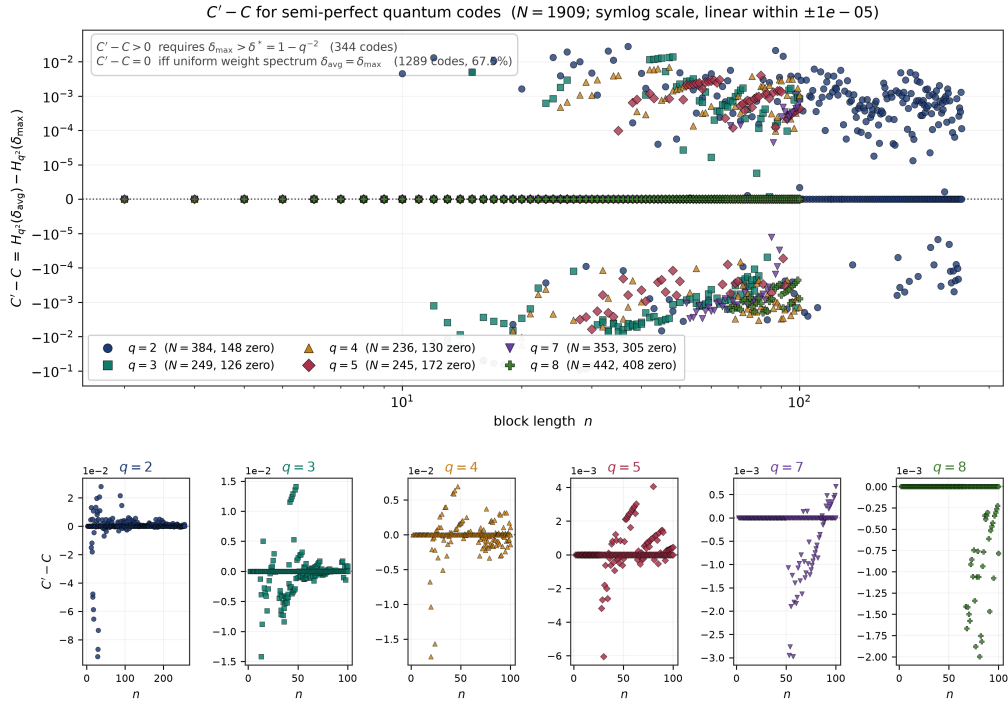


Figure 2.5.1: Comparison of C' and C as functions of block length n for semi-perfect quantum codes.

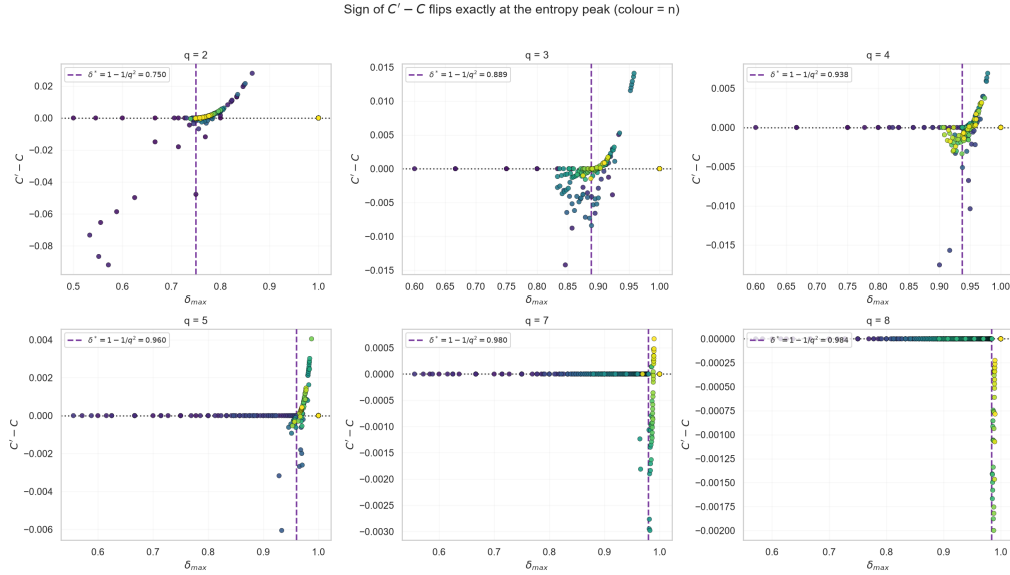


Figure 2.5.2: Sign of $C' - C$ as a function of δ_{max} for semi-perfect quantum codes.

2.6 Predictability of the Correction Factors

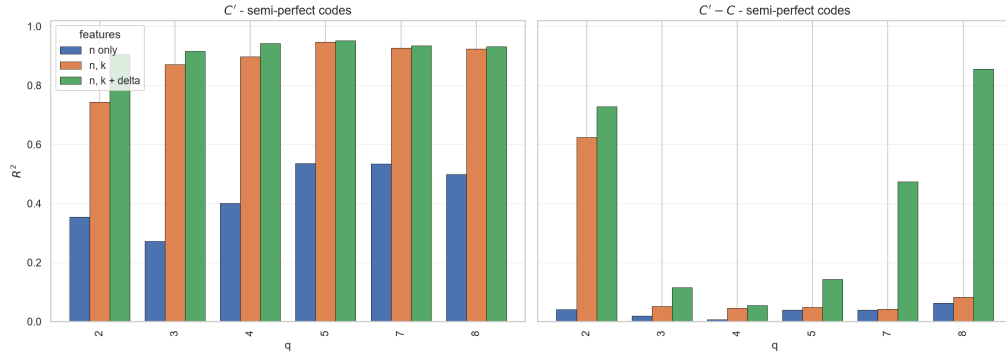


Figure 2.6.1: R^2 values for predicting C' from different feature sets for semi-perfect quantum codes.

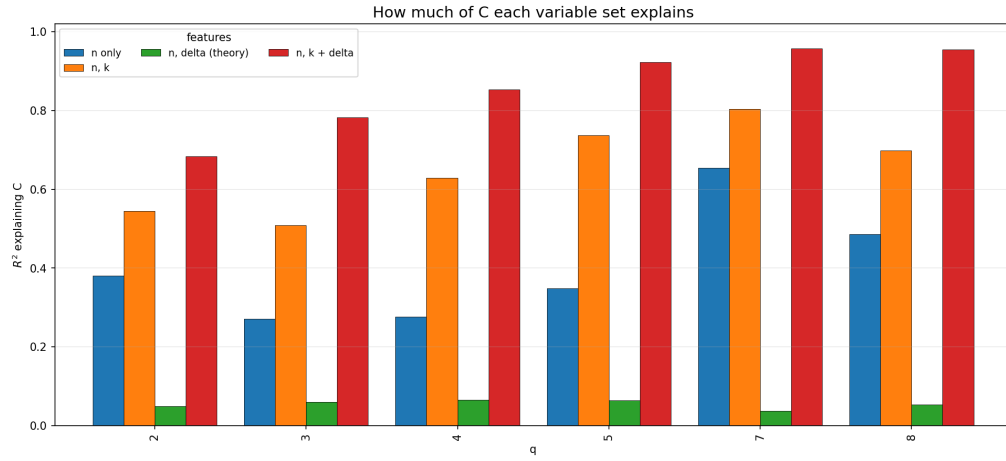


Figure 2.6.2: Comparison of the explanatory power of different feature sets for C .